Nome _____ N.º Aluno _____

Curso AULA 2 Ano Letivo / Data da Avaliação / /

Prova Escrita de: _____ N.º Folhas _____

Época: _____

Prefixos:

$1m = 10^{-3}$ (mili)	$1k = 10^{3}$ (kilo)
$1\mu = 10^{-6}$ (micro)	$1M = 10^{6}$ (Mega)
$1n = 10^{-9}$ (nano)	$1G = 10^{9}$ (Giga)
$1p = 10^{-12}$ (pico)	$1T = 10^{12}$ (Tera)

Lei do OHM:

$V = R \times I$

V – Tensão eléctrica (Volt - V)

R – Resistência eléctrica (ohm - Ω)

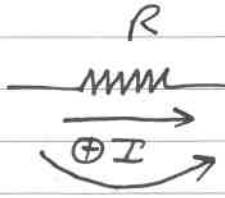
I – corrente eléctrica (Ampere - A)

Potência eléctrica:

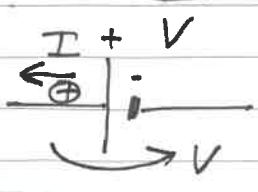
P → Potência eléctrica (watt - W)

$P = V \times I$

(convenção passiva)



P > 0

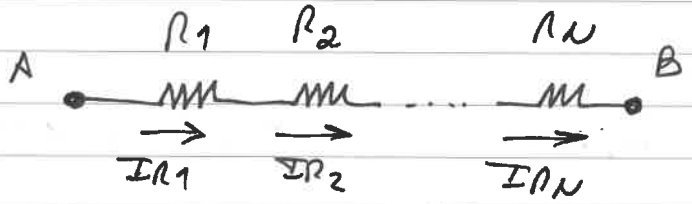


P < 0

RESISTÊNCIAS

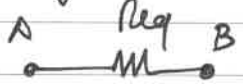
- SÉRIE → corrente comum.
- PARALELO → queda de tensão comum.

• SÉRIE

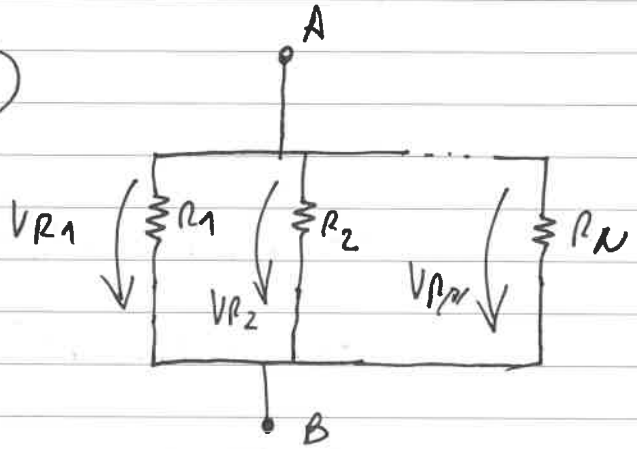


$$I_{R1} = I_{R2} = \dots = I_{RN}$$

$$R_{eq} = \sum_{i=1}^{N} R_i$$

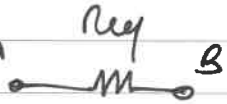


• PARALELO



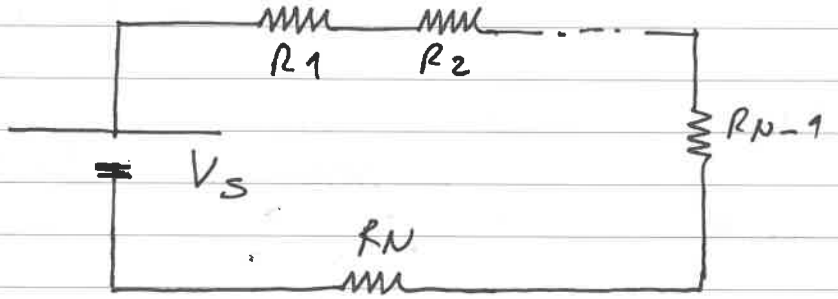
$$V_{R1} = V_{R2} = \dots = V_{RN}$$

$$\frac{1}{R_{eq}} = \sum_{i=1}^{N} \frac{1}{R_i} \Rightarrow$$



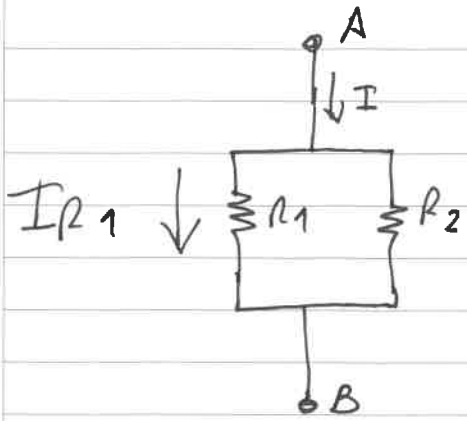
---

1) DIVISOR DE TENSÃO



$$V_{Ri} = \frac{R_i \times V_s}{\sum_{j=1}^{N} R_j}$$

Divisor de corrente:



$$I_{R_1} = \frac{R_2 \times I}{R_1 + R_2}$$

Exercícios:

1)

a) $I = 10\,mA$ $\qquad 1m = 10^{-3}$

$I = 10 \times \underbrace{10^{-3}}_{m}\ A = 10^{-2}\ A = 0{,}01\ A$

b) $I = 0{,}01\ mA$ $\qquad 1\mu = 10^{-6}$

$I = 0{,}01 \times 10^{-3}\ A = 10^{-5}\ A = 10 \times \underbrace{10^{-6}}_{\mu}\ A = 10\,\mu A$

c) $V = 100\,V$ $\qquad 1k = 10^{3}$

$V = 100\,V = 10^{2}\,V = 0{,}1 \times 10^{3}\,V = 0{,}1\,kV$

d) $P = 0{,}1\,MW$ $\qquad 1M = 10^{6}$

$P = 0{,}1 \times 10^{6}\,W = 10^{5}\,W = 10^{2} \times 10^{3}\,W = 100\,kW$

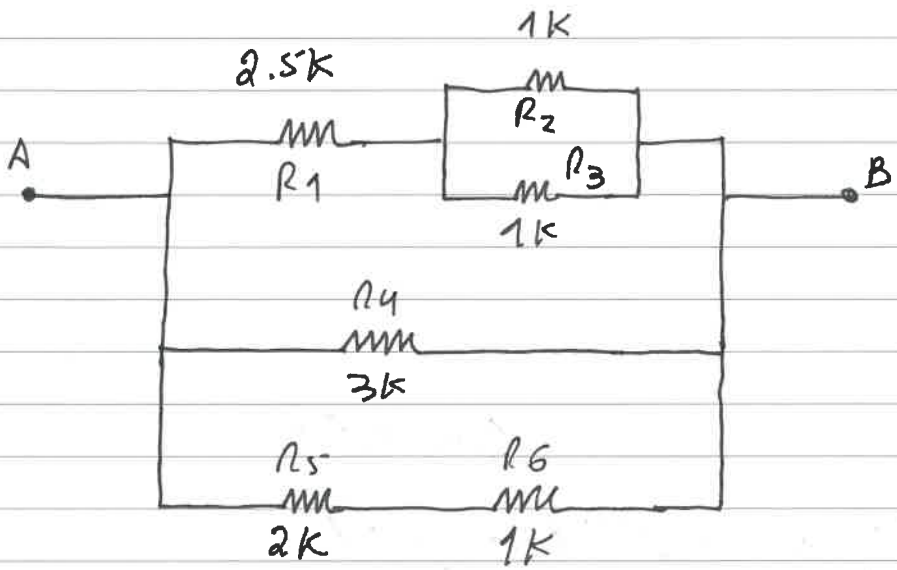
c) $\epsilon = 90.25\ MJ$ (4)

$\epsilon = 90.25 \times 10^6\ J = 90.25 \times 10^3 \times 10^3\ J$

$\epsilon = 90.25 \times 10^3\ kJ = 25\ kJ$

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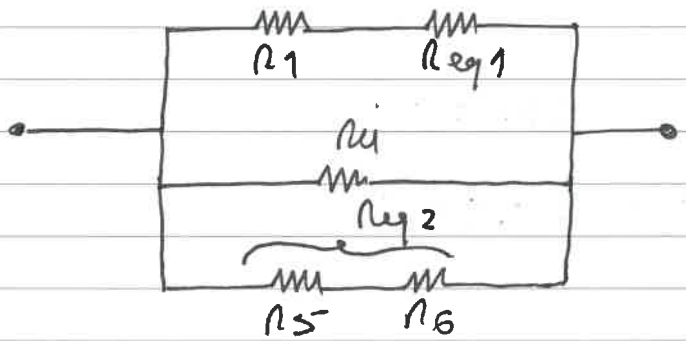
2)



a) SERIE = { R5, R6 }

b) PARALLELO = { R2, R3 }

c)



$Req_1 = R_2 // R_3$

$Req_2 = R_5 + R_6$

$= 2K + 1K = 3K$

$\frac{1}{Req_1} = \frac{1}{1K} + \frac{1}{1K} = \frac{2}{1K}$

$Req_1 = \frac{1K}{2} = 500\Omega$

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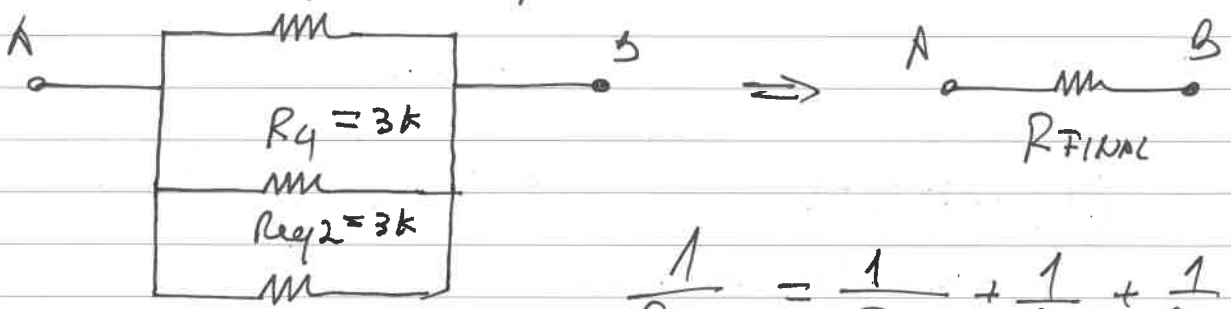
Curso _____ Ano Letivo ____ / ____ Data da Avaliação ____ / ____ / ____

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2c) (continuação)

$R_{eq3} = R1 + R_{eq1} = 3k$ (2.5k + 0.5k)



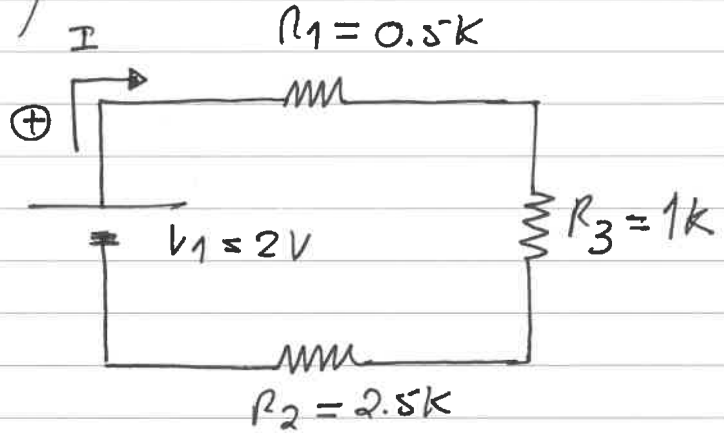
$$\frac{1}{R_{FINAL}} = \frac{1}{R_{eq3}} + \frac{1}{R_4} + \frac{1}{R_{eq2}}$$

$$\frac{1}{R_{FINAL}} = \frac{1}{3k} + \frac{1}{3k} + \frac{1}{3k} = \frac{3}{3k}$$

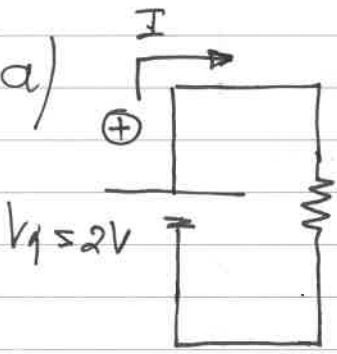
$$R_{FINAL} = \frac{3k}{3} = 1k\Omega.$$

- EXPLICAR CÓDIGO DO COMGS ⇒ VISITAR SITE
- MOSTRAR SIMULAÇÃO TINKERCAD

3)



a)

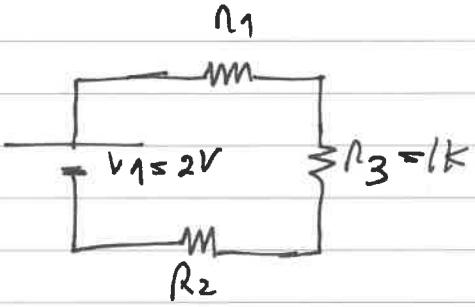


$= 0.5k + 1k + 2.5k = 4k$

$V = RI$

$I = \frac{V}{R} = \frac{2}{4k} = 0,5mA$

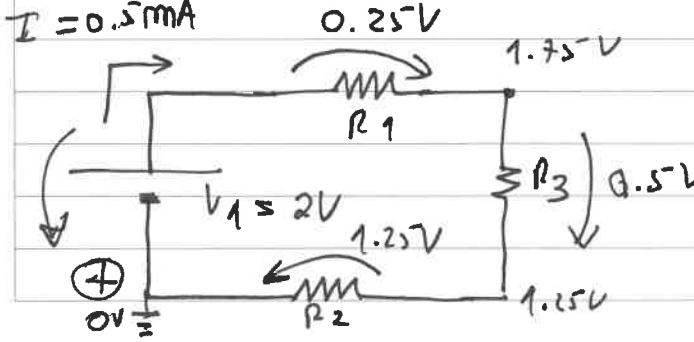
b) $V_{R_3} = ?$



$V_{R_3} = R_3 \times I = 1k \times 0.5m = \underline{\underline{0.5V}}$

c)

$V_{R_1} = \frac{R_1 \times V_1}{R_1 + R_2 + R_3} = \frac{0.5k \times 2}{0.5k + 2.5k + 1k} = \frac{0.5k \times 2}{4k} = 0,25V$



$V_{R_2} = R_2 \times I = 2.5k \times 0.5m$

$= 1.25V$

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d)

$P = \frac{\epsilon}{\Delta T}$ $\quad \Delta T = 30 \text{ dias}$

$\epsilon = P \times \Delta T$ $\quad \Delta T = 30 \times 24 \times 60 \times 60 =$

$P = V \times I = 2 \times 0.5m = 1 mW$

$\epsilon = 1(mW) \times \Delta T = 10^{-3}(W) \times \Delta T = 10^{-3}(J/s) \times 30 \times 24 \times 60 \times 60(s)$

J/s

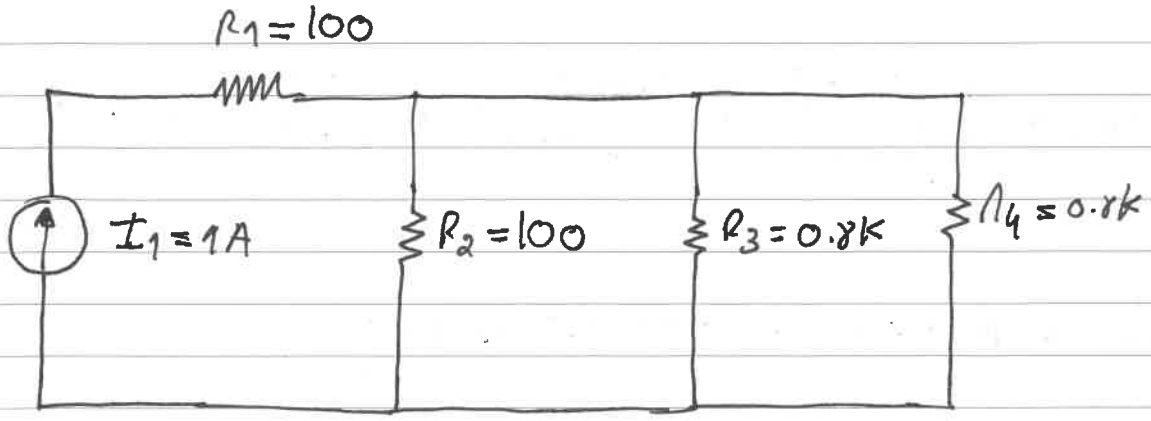
$\epsilon = 10^{-3} \times 30 \times 24 \times 60 \times 60 = 2592 J \cong 2{,}5 kJ.$

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e) FAZER SIMULAÇÕES | PSPICE
CTSPICE

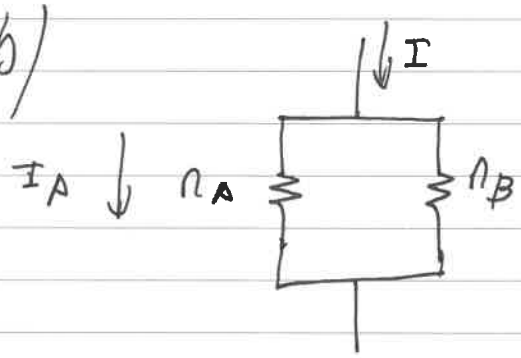
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4)

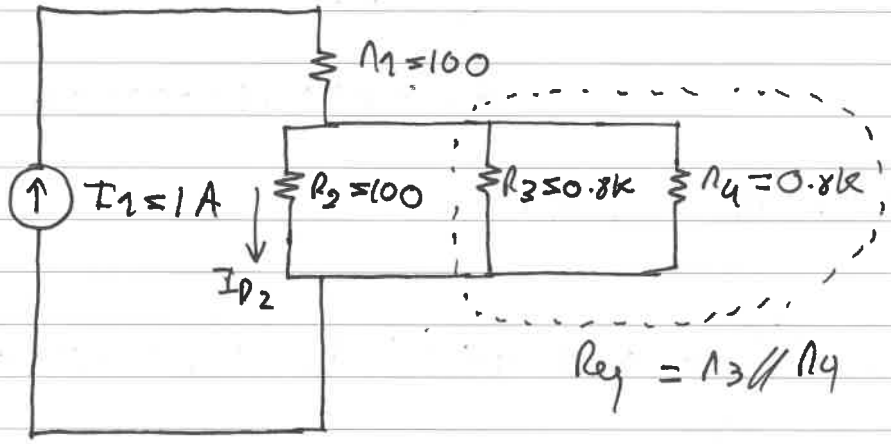


a) $I_{R1} = I_1 = 1A.$

b)

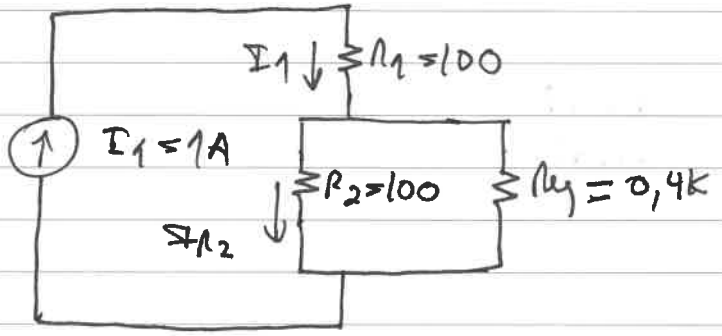


$$I_A = \frac{R_B \times I}{R_A + R_B}$$



$$\frac{1}{R_{eq}} = \frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{0.8k} + \frac{1}{0.8k} = \frac{2}{0.8k}$$

$$R_{eq} = \frac{0.8k}{2} = 0.4k$$



$$I_{R_2} = \frac{R_{eq} \times I_1}{R_{eq} + R_2} = \frac{0.4k \times 1}{0.4k + 100} = \frac{400}{500} = \underline{0.8A}$$

c)

$$V_{R_2} = R_2 \times I_{R_2} = 100 \times 0.8 = \underline{80V}$$

d)

$$P = -V_I \times I$$

$$= -V_I \times 1 \quad = -180 \times 1 = -180 \text{ watts}$$

$$V_I = V_{R1} + V_{R2} = 1 \times 100 + 80 = 180V$$

e) $P = \frac{\epsilon}{\Delta T}$

$$\epsilon = P \times \Delta T = \underset{(J/s)}{180} \times \underset{(s)}{120 \times 24 \times 60 \times 60} \; J$$

$$\rightarrow \epsilon_J = 1866240000 \; J = 1866240 \; kJ$$

$$1 \text{ cal} \; - \; 4.187 \; J$$

$$\underset{\epsilon \text{ cal}}{x} \; - \; 1866240000 \; J$$

$$\rightarrow \epsilon_{cal} = \frac{1866240000}{4.187} = 445722474 \text{ cal}$$

$$\simeq 445723 \text{ kcal}$$

$$\rightarrow \epsilon_{wh} = \underset{(w)}{180} \times \Delta T = 180 \times 120 \times 24 = 518400 \text{ wh}$$

$$\epsilon_{kwh} = 518.4 \text{ kwh}$$

(10)

d)

0.25 Euros / kwh.

$\Delta T = 1$ mes

$\epsilon = P \times \Delta T = 180 \times 30 \times 24 = 129600$ wh.

$\epsilon = 129.6$ kwh

CUSTO

1 kwh — 0,25 EUROS

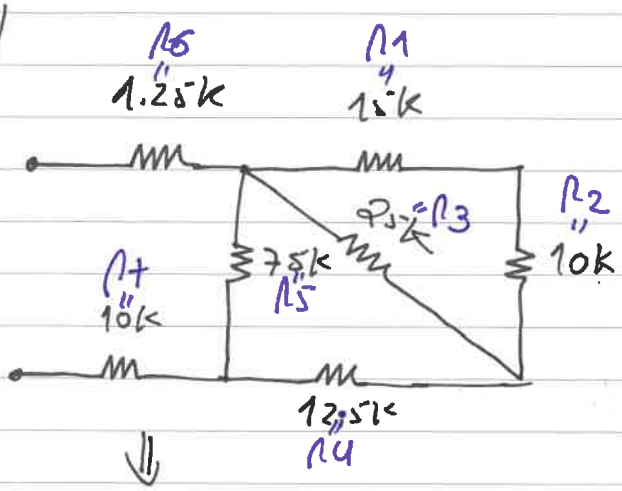
129.6 kwh — x.

$x = 0,25$ EUROS $\times$ 129.6 kwh = 32.4 EUROS

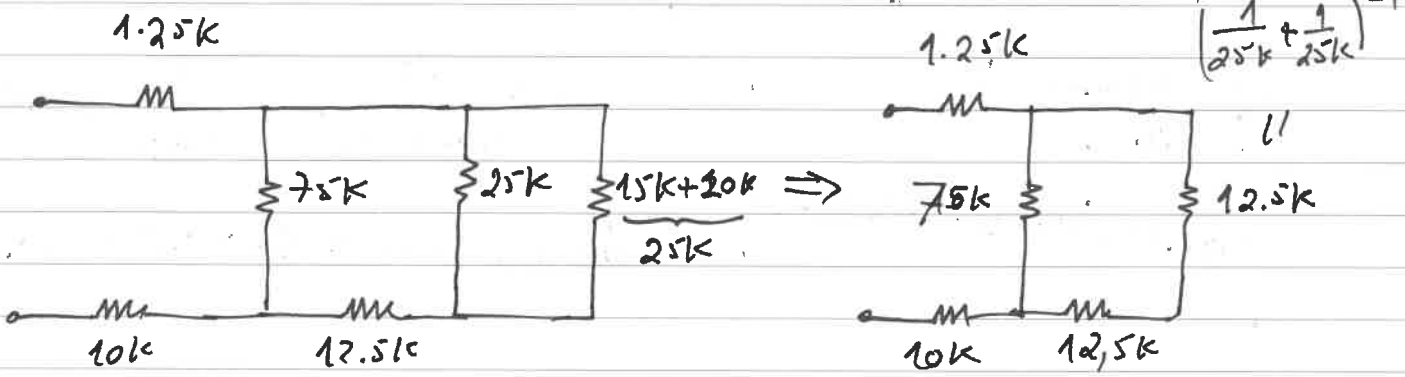
31
28
31
30
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30
31
31
30
31
30
31

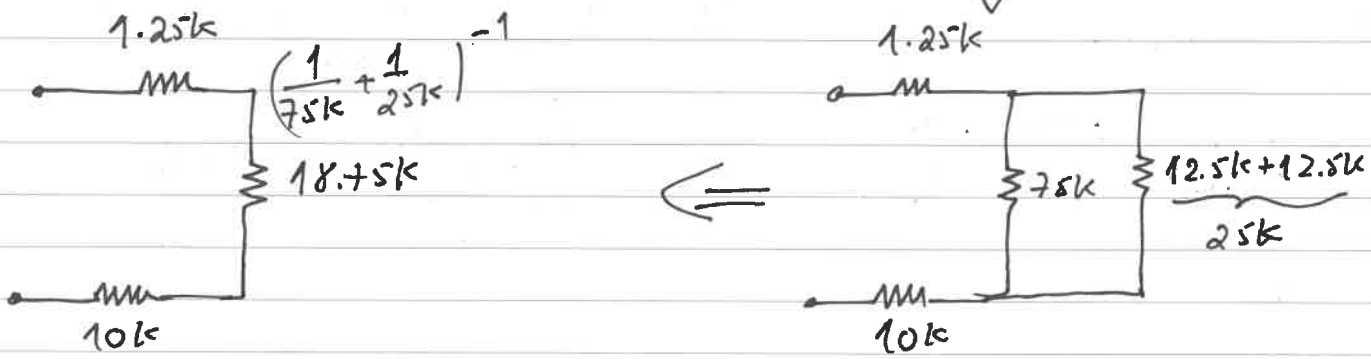
$\frac{365 \text{ dias}}{12} = 30.416$

5/



$R_1 = 15k$
$R_2 = 10k$
$R_3 = 25k$
$R_4 = 12.5k$
$R_5 = 7.5k$
$R_6 = 1.25k$
$R_7 = 10k$



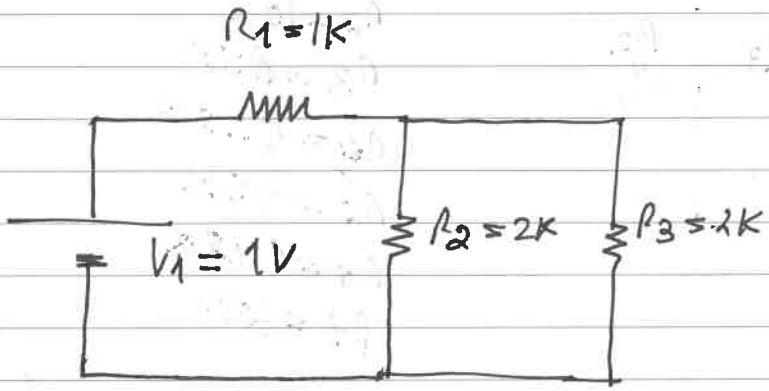


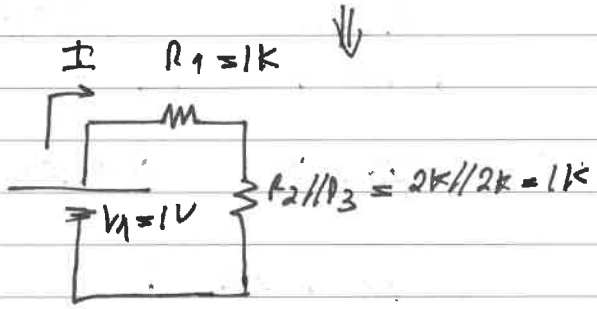
$$Req_{FINAL} = 1.25k + 18.75k + 10k = 30k\Omega$$



6)

a)

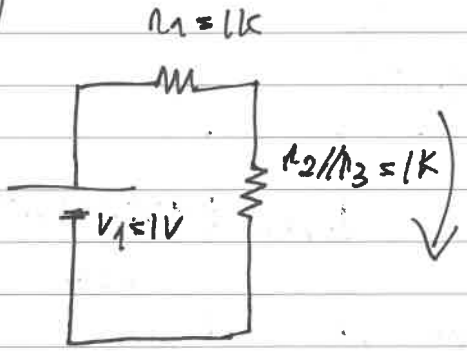




$$I = \frac{V_1}{R_1 + R_2 // R_3} = \frac{1}{1k+1k} = 0.5mA$$

$$V_{R_1} = R_1 \times I = 1k \times 0.5m = 0.5V$$

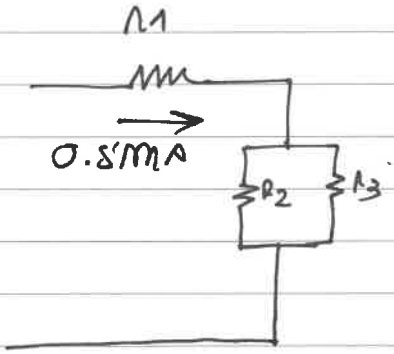
b)



$$V_{R_3} = V_{R_2 // R_3} = \frac{1k \times 1}{1k+1k} = 0.5V$$

$$I_{R_3} = \frac{V_{R_3}}{R_3} = \frac{0.5}{2k} = 0.25mA$$

b)



$$I_{R_2} = \frac{R_2 \times I}{R_2 + R_3} = \frac{1k \times 0.5m}{1k+1k} = 0.25mA$$

$$V_{R_3} = R_3 \times I_{R_3} = 2k \times 0,25mA = 0.5V$$

d) $P_{R_1} = R_1 I_{R_1}^2 = 1k \times (0.5m)^2 = 0.00025W = 0.25mW$

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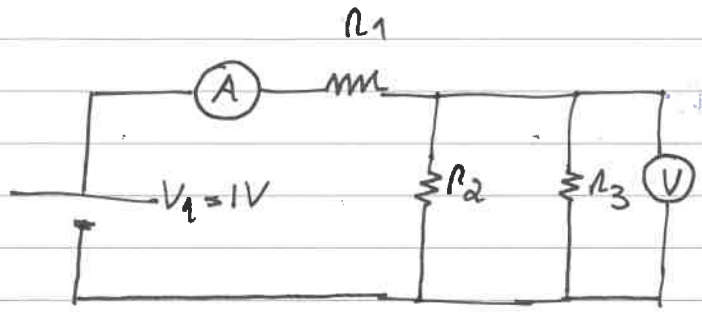
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e)



f) $P = V \times I$

$P_V = -1 \times 0.5m = -0.5mW$. (FONTE)

$P_{R1} = R1 \times I_{R1}^2 = 0.25mW = 0.25mW$

$P_{R2} = R_2 \times I_{R_2}^2 = 2k \times (0.25m)^2 = 0.125mW$

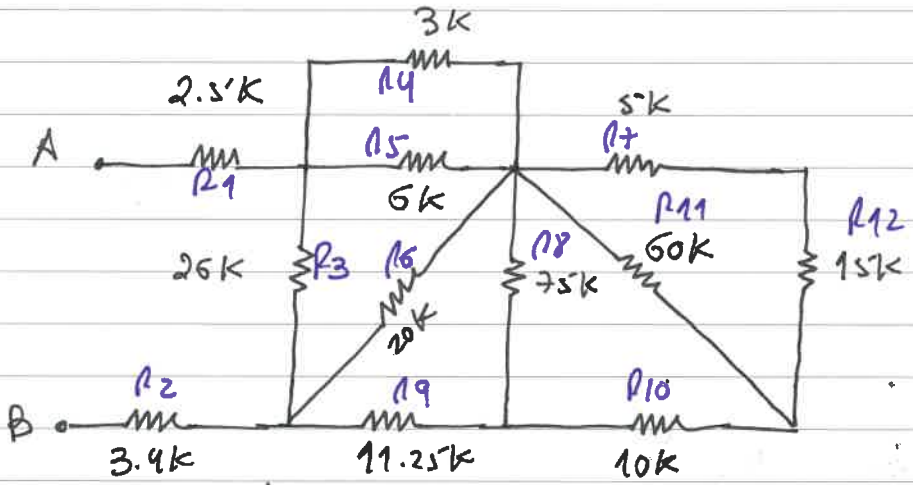
$P_{R3} = R3 \times I_{R3}^2 = 2k \times (0.25m)^2 = 0.125mW$

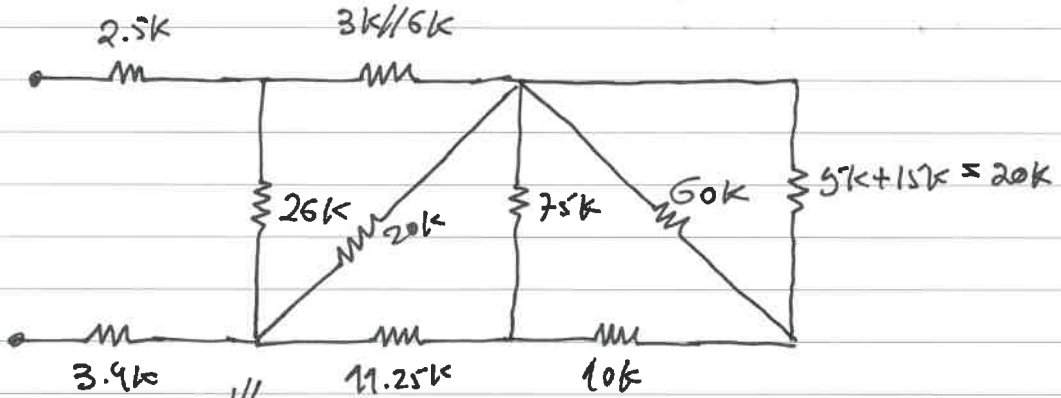
TOTAL CONSUMIDO = 0.5mW

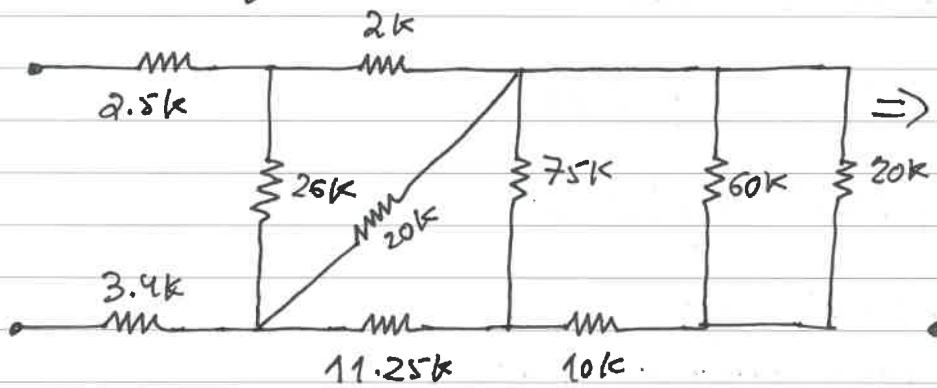
$0.5mW = 0.25mW + 0.125mW + 0.125mW$

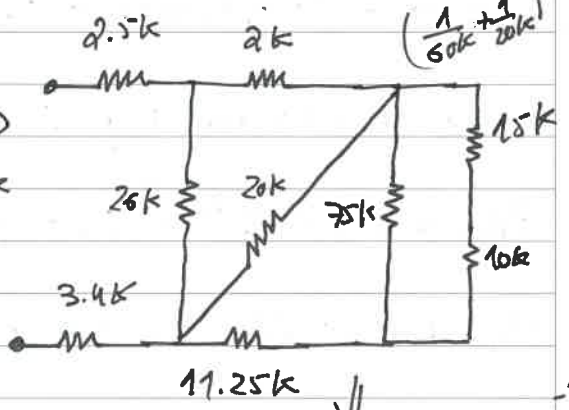
FORNECIDO — consumido R1 — consumido R2 — consumido R3

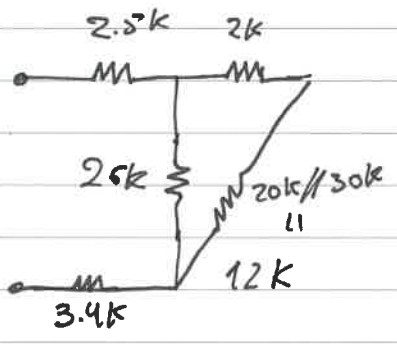
7)

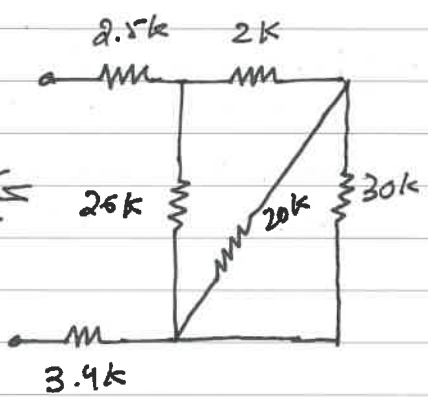

3K
2.5K
R4
A
R1
R5
6K
R7
5K
R11
60K
R12
15K
26K
R3
R6
20K
R8
75K
R2
R9
R10
B
3.4K
11.25K
10K

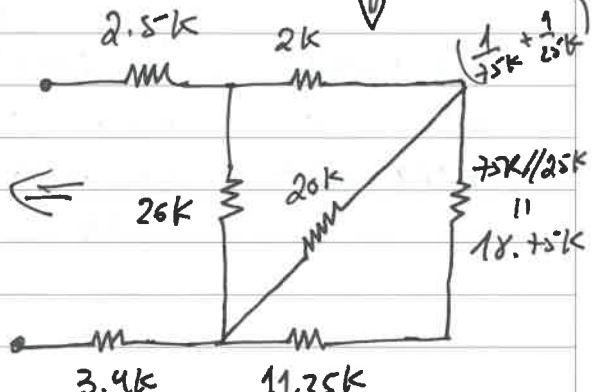

2.5K
3K//6K
26K
20K
75K
60K
5K+15K = 20K
3.4K
11.25K
10K

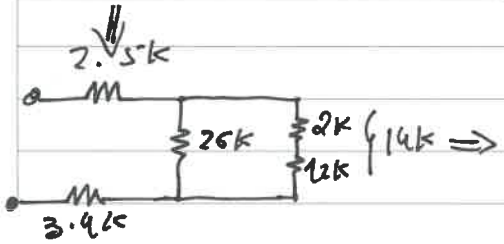

2K
2.5K
26K
20K
75K
60K
20K
3.4K
11.25K
10K

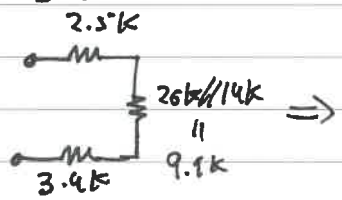

2.5K
2K
(1/60K + 1/20K)^-1
15K
26K
20K
75K
10K
3.4K
11.25K


2.5K
2K
26K
20K//30K
12K
3.4K


2.5K
2K
26K
20K
30K
3.4K


2.5K
2K
(1/75K + 1/25K)^-1
26K
20K
75K//25K
18.75K
3.4K
11.25K


2.5K
26K
2K
12K
14K
3.4K


2.5K
26K//14K
9.1K
3.4K


2.5K
9.1K
3.4K
15K